Exp No:4

To Count the frequency of occurrence of a word in a body of text is often needed during text processing.

Aim :

To Import the word_tokenize function and gutenberg.

Code :

```
import nltk
from nltk.tokenize import word_tokenize
from nltk.corpus import gutenberg
# Download the gutenberg corpus if not already installed
nltk.download('gutenberg')
nltk.download('punkt') # Ensure the punkt tokenizer models are also downloaded
# Load the text from the gutenberg corpus
sample = gutenberg.raw("austen-emma.txt") # Change to an existing text from the corpus
# Tokenize the sample text
token = word_tokenize(sample)
# Create a list of the first 50 tokens
wlist = []
for i in range(50):
wlist.append(token[i])
# Calculate the frequency of each word in the list
wordfreq = [wlist.count(w) for w in wlist]
# Print the word-frequency pairs
print("Pairs\n" + str(list(zip(wlist, wordfreq))))
```

OUTPUT:

Pairs

[('[', 1), ('Emma', 2), ('by', 1), ('Jane', 1), ('Austen', 1), ('1816',

1), (']', 1), ('VOLUME', 1), ('I', 2), ('CHAPTER', 1), ('I', 2), ('Emma

', 2), ('Woodhouse', 1), (',', 5), ('handsome', 1), (',', 5), ('clever'

, 1), (',', 5), ('and', 3), ('rich', 1), (',', 5), ('with', 2), ('a', 1

), ('comfortable', 1), ('home', 1), ('and', 3), ('happy', 1), ('disposi

tion', 1), (',', 5), ('seemed', 1), ('to', 1), ('unite', 1), ('some', 1

), ('of', 2), ('the', 2), ('best', 1), ('blessings', 1), ('of', 2), ('e

xistence', 1), (';', 1), ('and', 3), ('had', 1), ('lived', 1), ('nearly

', 1), ('twenty-one', 1), ('years', 1), ('in', 1), ('the', 2), ('world'

, 1), ('with', 2)]

Result :

Thus the frequency of occurrence of a word in a body of text is counted.